Pizza Sales Analysis Using SQL

Analysing Sales Performance and Customer Ordering Patterns

Tool Used : MySQL | Excel | PowerPoint

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PROJECT OVERVIEW

To analyse pizza sales data and uncover business insights using SQL queries.

Dataset Tables:
- Orders
- Order Details
- Pizzas
- Pizza Types

Key Goals:
- Revenue Analysis
- Customer Preferences
- Sales Trends
- Product Performance

Retrieve the total number of orders placed.

```
select count(*) as total_orders
from orders;
```

total_orders
21350

Calculate the total revenue generated from pizza sales.

	total_sales
▶	800546.2

```
SELECT
    ROUND(SUM(order_details.quantity * pizzas.price),
            2) AS total_sales
FROM
    order_details
        JOIN
    pizzas ON pizzas.pizza_id = order_details.pizza_id
```

Identify the highest-priced pizza.

```sql
SELECT
    pizza_types.name, pizzas.price
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1;
```

name	price
The Greek Pizza	35.95

Identify the most common pizza size ordered.

```
SELECT
    pizzas.size,
    COUNT(order_details.order_details_id) AS order_count
FROM
    pizzas
        JOIN
    order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizzas.size
ORDER BY order_count DESC;
```

size	order_count
L	18124
M	15064
S	13848
XL	535
XXL	28

List the top 5 most ordered pizza types along with their quantities.

name	quantity
The Classic Deluxe Pizza	2390
The Barbecue Chicken Pizza	2383
The Pepperoni Pizza	2373
The Hawaiian Pizza	2364
The California Chicken Pizza	2321

```
SELECT
    pizza_types.name, SUM(order_details.quantity) AS quantity
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC
LIMIT 5;
```

Join the necessary tables to find the total quantity of each pizza category ordered.

```sql
SELECT
    pizza_types.category,
    SUM(order_details.quantity) AS quantity
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC;
```

category	quantity
Classic	14552
Supreme	11766
Veggie	11385
Chicken	10820

Determine the distribution of orders by hour of the day.

```sql
select hour(order_time) as hour,
count(order_id) as order_count
from orders
group by hour(order_time);
```

hour	order_count
15	1468
16	1920
17	2336
18	2399
19	2009
20	1642
21	1198
22	663
23	28
10	8
9	1

Join relevant tables to find the category-wise distribution of pizzas.

```
select category , count(name) from pizza_types
group by category
```

category	count(name)
Chicken	6
Classic	8
Supreme	9
Veggie	9

Group the orders by date and calculate the average number of pizzas ordered per day.

```sql
SELECT
    ROUND(AVG(quantity), 0) as avg_pizza_order_per_day
FROM
    (SELECT
        orders.order_date, SUM(order_details.quantity) AS quantity
    FROM
        orders
    JOIN order_details ON orders.order_id = order_details.order_id
    GROUP BY orders.order_date) AS order_quantity;
```

avg_pizza_order_per_day
139

```sql
select pizza_types.name,
sum(order_details.quantity * pizzas.price) as revenue
from pizza_types join pizzas
on pizzas.pizza_type_id = pizza_types.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.name order by revenue desc limit 3;
```

Determine the top 3 most ordered pizza types based on revenue.

name	revenue
The Thai Chicken Pizza	42397
The Barbecue Chicken Pizza	41899.25
The California Chicken Pizza	40564.75

Calculate the percentage contribution of each pizza type to total revenue.

category	revenue
Classic	26.87
Supreme	25.53
Chicken	23.96
Veggie	23.64

```
select pizza_types.category,
round(sum(order_details.quantity*pizzas.price) / (select
    ROUND(SUM(order_details.quantity * pizzas.price),
            2) AS total_sales
FROM
    order_details
        JOIN
    pizzas ON pizzas.pizza_id = order_details.pizza_id) * 100,2 )as revenue
from pizza_types join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.category order by revenue desc;
```

Analyze the cumulative revenue generated over time.

order_date	cum_revenue
2015-01-01	2713.8500000000004
2015-01-02	5445.75
2015-01-03	8108.15
2015-01-04	9863.6
2015-01-05	11929.55
2015-01-06	14358.5
2015-01-07	16560.7
2015-01-08	19399.05
2015-01-09	21526.4
2015-01-10	23990.350000000002
2015-01-11	25862.65
2015-01-12	27781.7
2015-01-13	29831.300000000003
2015-01-14	32358.700000000004
2015-01-15	34343.50000000001

```
select order_date,
sum(revenue) over(order by order_date) as cum_revenue
from
(select orders.order_date,
sum(order_details.quantity * pizzas.price) as revenue
from order_details join pizzas
on order_details.pizza_id = pizzas.pizza_id
join orders
on orders.order_id = order_details.order_id
group by orders.order_date) as sales;
```

Determine the top 3 most ordered pizza types based on revenue for each pizza category.

name	revenue
The Thai Chicken Pizza	42397
The Barbecue Chicken Pizza	41899.25
The California Chicken Pizza	40564.75
The Classic Deluxe Pizza	37190
The Hawaiian Pizza	31486.75
The Pepperoni Pizza	29604.75
The Spicy Italian Pizza	34138.5
The Italian Supreme Pizza	32942.5
The Sicilian Pizza	30319.5
The Four Cheese Pizza	31468.8000000062
The Mexicana Pizza	26227.75
The Five Cheese Pizza	25567

```
select name, revenue from
(select category, name, revenue,
rank() over( partition by category order by revenue desc) as rn
from
(select pizza_types.category, pizza_types.name,
sum((order_details.quantity) * pizzas.price) as revenue
from pizza_types join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.category, pizza_types.name) as a) as b
where rn <=3;
```